Lab 01

Course: Data Structure and algorithm

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1. Write a C++ program to ask information from a student such as name, gender, major, age then display the greeting message on screen. Say he/she is eligible to vote the election if the age is at least 18.

   Hi, Mr. ***name!*** your age is ***age*** year old and you learn ***major***! : **for male (M)**

   Hi, Mrs. n***ame!***, you are ***age*** year old and you are majoring in ***major*** : **for female (F)**

   For either Male or Female:

   You can vote. If age is at least 18.

   You can not vote Otherwise.

The code:

```cpp
#include <iostream>
#include <string>
using namespace std;
int main(){
    string name;
    int age;
    string major;
    string gender;

    cout << "Enter your name: ";
    cin >> name;
    cout << "Enter your age: ";
    cin >> age;
    cout << "Gender: ";
    cin >> gender;
    cout << "Enter your major: ";
    cin >> major;
    cout << endl;

    if (gender == "F" || gender == "f"){
        cout << "Hi, Mrs. " << name << "!, you are " << age << " years old and you are majoring in " << major << endl;
    }
    else if (gender == "M" || gender == "m"){
        cout << "Hi, Mr. " << name << "!, you are " << age << " years old and you learn " << major << endl;
    }

    if (age < 18){
        cout << "You cannot vote" << endl;
    }
    else {
        cout << "You can vote" << endl;
    }
}
```

The output:

```
Enter your name: lyda
Enter your age: 20
Gender: f
Enter your major: SE

Hi, Mrs. lyda!, you are 20 years old and you are majoring in SE
You can vote
```

2. Write a C++ program to check whether an input character is a vowel or a consonant. Hint: Use ASCII code to test condition.
    - Uppercase letters from: 65 to 90, lowercase letters from 97 to 122
    - Vowel and its ASCII code: a = 97, e = 101, i = 105, o = 111, u = 117

The code:

```cpp
#include <iostream>
#include <string>
using namespace std;
int main (){
    char c;
    int ascii;
    cout << "Enter a character: ";
    cin >> c;
    ascii = (int)c;

    if ((ascii >= 65 && ascii <= 90) || (ascii >= 97 && ascii <= 122)) {
        if (ascii == 65 || ascii == 69 || ascii == 73 || ascii == 79 || ascii == 85 ||
            ascii == 97 || ascii == 101 || ascii == 105 || ascii == 111 || ascii == 117)
        {
            cout << c << " is a vowel." << endl;
        }
        else {
            cout << c << " is a consonant." << endl;
        }
    }
}
```

The output:

```
Enter a character: a
a is a vowel.
```

3. A program to compute tax salary of a person. The program asks for name, gender and salary of a person and tell him/her how much tax he/she is required to pay. The tax is computed based on the rules below:
   - **For males**
     - Salary more than 1000 USD, pay tax 9.5%
     - Salary 500 – 1000 USD, pay tax 7%
     - Salary 300 – 500 USD, pay tax 5%
     - Salary less than 300, no need to pay tax
   - **For females**
     - Salary more than 1000 USD, pay tax 8%
     - Salary 500 – 1000 USD, pay tax 6.5%
     - Salary 300 – 500 USD, pay tax 3.5%
     - Salary less than 300, no need to pay tax

The code:

```cpp
#include <iostream>
#include <string>
using namespace std;
int main (){
    string name;
    string gender;
    int salary;

    cout << "Enter your name: ";
    cin >> name;
    cout << endl;
    cout << "Gender: ";
    cin >> gender;
    cout << endl;
    cout << "Enter your salary: ";
    cin >> salary;
    cout << endl;

    if (gender == "female" || gender == "f"){
        if (salary < 300){
            cout << "No need to pay tax!" << endl;
        }
        else if (salary >= 300 && salary < 500){
            cout << "You have to pay 3.5% tax" << endl;
            cout << "Your tax is: " << salary * 0.035 << endl;
        }
        else if (salary >= 500 && salary < 1000){
            cout << "You need to pay 6.5% tax!" << endl;
            cout << "Your tax is: " << salary * 0.065 << endl;
        }
        else if (salary >= 1000){
            cout << "You need to pay 8% tax!" << endl;
            cout << "Your tax is: " << salary * 0.08 << endl;
        }
    }
```

```cpp

    if (gender == "male" || gender == "m"){
        if (salary < 300){
            cout << "No need to pay tax!" << endl;
        }
        else if (salary >= 300 && salary < 500){
            cout << "You have to pay 5% tax" << endl;
            cout << "Your tax is: " << salary * 0.05 << endl;
        }
        else if (salary >= 500 && salary < 1000){
            cout << "You need to pay 7% tax!" << endl;
            cout << "Your tax is: " << salary * 0.07 << endl;
        }
        else if (salary >= 1000){
            cout << "You need to pay 9.5% tax!" << endl;
            cout << "Your tax is: " << salary * 0.095 << endl;
        }
    }
}
```

The output:

```
Enter your name: lyda

Gender: f

Enter your salary: 1234

You need to pay 8% tax!
Your tax is: 98.72
```

4. Create a C++ program that can play a game Rock-Paper-Scissor. The game plays between the user against the computer.

The code:

```cpp
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;

int main() {
    srand(time(0));

    int userChoice, computerChoice;

    cout << "Let's play rock, paper, scissors! " << endl;
    cout << "1. Rock" << endl;
    cout << "2. Paper" << endl;
    cout << "3. Scissors" << endl;
    cout << "Enter your choice (1-3): ";
    cin >> userChoice;

    if (userChoice < 1 || userChoice > 3) {
        cout << "Invalid choice! Choose a number from 1-3" << endl;
        return 0;
    }

    computerChoice = (rand() % 3) + 1;

    string choices[] = {"", "Rock", "Paper", "Scissors"};
    cout << "You chose: " << choices[userChoice] << endl;
    cout << "Computer chose: " << choices[computerChoice] << endl;

    if (userChoice == computerChoice) {
        cout << "It's a tie, GG" << endl;
    }
    else if ((userChoice == 1 && computerChoice == 3) ||
             (userChoice == 2 && computerChoice == 1) ||
             (userChoice == 3 && computerChoice == 2)) {
        cout << "You won! YAY" << endl;
    }
    else {
        cout << "I WIN! HAHA " << endl;
    }

    return 0;
}
```

The output:

```
Let's play rock, paper, scissors!
1. Rock
2. Paper
3. Scissors
Enter your choice (1-3): 2
You chose: Paper
Computer chose: Paper
It's a tie, GG
```

5. Write a C++ program to convert a minute to a time format which consists of hour, minute, and second (h:m:s). A user is required to input a minute.
    - Hours = minutes / 60
    - Remainderminutes = minutes % 60
    - Seconds = remainderminutes * 60

The code:

```cpp
#include <iostream>
using namespace std;
int main(){
    int min;
    cout << "Enter the minute you want to convert: ";
    cin >> min;

    int hour = min / 60;
    int remainMin = min % 60;
    int sec = remainMin * 60;

    cout << endl;
    cout << min << " minute(s) is equal to " << hour << ": " << remainMin << ": " << sec << endl;
}
```

The output:

```
Enter the minute you want to convert: 90

90 minute(s) is equal to 1: 30: 1800
```

6. Write a C++ program to find the summation of numbers from 1 to n except number 10 and 30, where n is a number input by a user and n should be greater than 50.

The code:

```cpp
#include <iostream>
using namespace std;

int main() {
    int n;
    int sum = 0;
    cout << "Enter a number that is greater than 50: ";
    cin >> n;

    if (n <= 50 ){
        cout << "Invalid number, enter one that is greater than 50" << endl;
        return 0;
    }
    for (int i = 1; i <=n ; i++){
        if (i == 10 || i == 30){
            continue;
        }
        sum += i;
    }
    cout << "The sum of the numbers from 1 to " << n << " is: " << sum << endl;

}
```

The output:

```
Enter a number that is greater than 50: 60
The sum of the numbers from 1 to 60 is: 1790
```

7. Write a C++ program to create a new data structure for storing info of book (book ID, book ISBN, book title, published year, author names and price ($)). Each book could have more than one author. To do:
   - Create an array that can store 5 books' info.
   - Create a function to display a book info based on ISBN. This function takes a parameter which is an ISBN of a book.

     void displayBookByISBN(Book books[], int size, string isbn){.... }
   - Create a function to display information of all books.

     void displayAllBooks(Book books[], int size) { ... }

The code:

```cpp
#include <iostream>
#include <string>
using namespace std;

struct Book {
    int bookID;
    string isbn;
    string title;
    int publishedYear;
    string authors[5];
    int authorCount;
    double price;
};

void displayBook(Book b) {
    cout << "Book ID:        " << b.bookID << endl;
    cout << "ISBN:           " << b.isbn << endl;
    cout << "Title:          " << b.title << endl;
    cout << "Published Year: " << b.publishedYear << endl;
    cout << "Authors:        ";
    for (int i = 0; i < b.authorCount; i++) {
        cout << b.authors[i];
        if (i < b.authorCount - 1) cout << ", ";
    }
    cout << endl;
    cout << "Price:          $" << b.price << endl << endl;
}

void displayBookByISBN(Book books[], int size, string isbn) {
    bool found = false;
    for (int i = 0; i < size; i++) {
        if (books[i].isbn == isbn) {
            displayBook(books[i]);
            found = true;
            break;
        }
    }
    if (!found) {
        cout << "Book with ISBN " << isbn << " not found." << endl;
    }
}
```

```cpp
void displayAllBooks(Book books[], int size) {
    cout << "=== All Books ===" << endl;
    for (int i = 0; i < size; i++) {
        displayBook(books[i]);
    }
}

int main() {
    Book books[5];

    books[0] = {1, "978-0-13-468599-1", "The C++ Programming Language", 2013,
                {"Bjarne Stroustrup"}, 1, 59.99};

    books[1] = {2, "978-0-13-235088-4", "Clean Code", 2008,
                {"Robert C. Martin"}, 1, 39.99};

    books[2] = {3, "978-0-20-163361-5", "The Pragmatic Programmer", 2019,
                {"David Thomas", "Andrew Hunt"}, 2, 49.99};

    books[3] = {4, "978-0-59-651798-1", "Head First Design Patterns", 2020,
                {"Eric Freeman", "Elisabeth Robson"}, 2, 54.99};

    books[4] = {5, "978-0-13-110362-7", "The C Programming Language", 1988,
                {"Brian Kernighan", "Dennis Ritchie"}, 2, 34.99};

    displayAllBooks(books, 5);


    cout << "\n=== Search by ISBN ===" << endl << endl;
    string searchISBN;
    cout << "Enter ISBN to search: ";
    cin >> searchISBN;
    displayBookByISBN(books, 5, searchISBN);

    return 0;
}
```

The output:

```
=== All Books ===
Book ID:        1
ISBN:           978-0-13-468599-1
Title:          The C++ Programming Language
Published Year: 2013
Authors:        Bjarne Stroustrup
Price:          $59.99

Book ID:        2
ISBN:           978-0-13-235088-4
Title:          Clean Code
Published Year: 2008
Authors:        Robert C. Martin
Price:          $39.99

Book ID:        3
ISBN:           978-0-20-163361-5
Title:          The Pragmatic Programmer
Published Year: 2019
Authors:        David Thomas, Andrew Hunt
Price:          $49.99

Book ID:        4
ISBN:           978-0-59-651798-1
ISBN:           978-0-13-235088-4
Title:          Clean Code
Published Year: 2008
Authors:        Robert C. Martin
Price:          $39.99

Book ID:        3
ISBN:           978-0-20-163361-5
Title:          The Pragmatic Programmer
Published Year: 2019
Authors:        David Thomas, Andrew Hunt
Price:          $49.99
```

```
Book ID:        4
ISBN:           978-0-59-651798-1
Title:          Head First Design Patterns
Published Year: 2020
Authors:        Eric Freeman, Elisabeth Robson
Price:          $54.99

Book ID:        5
ISBN:           978-0-13-110362-7
Title:          The C Programming Language
Published Year: 1988
Authors:        Brian Kernighan, Dennis Ritchie
Price:          $34.99


=== Search by ISBN ===

Enter ISBN to search: 978-0-20-163361-5
Book ID:        3
ISBN:           978-0-20-163361-5
Title:          The Pragmatic Programmer
Published Year: 2019
Authors:        David Thomas, Andrew Hunt
Price:          $49.99
```

8. Create 5 functions to:
   - i) convert temperature Celsius to Fahrenheit: Fahrenheit = (Celsius * 9.0 / 5.0) + 32
   - ii) convert temperature Fahrenheit to Celsius: Celsius = (Fahrenheit – 32) * 50 / 9.0
   - iii) find root of quadratic equation $ax^2+bx+c=0$.
   - iv) compute BMI of a person's weight and height and tell whether he/she is overweight, underweight and other terms according to BMI list:
     - bmi = weight / (height * height)
     - bmi < 18.5, underweight
     - bmi < 25, normal weight
     - bmi < 30, overweight
     - else, obese
   - v) sum numbers from 1 to n except those numbers that are divisible by 3, n is a parameter of the function.

   Design a menu program to demonstrate these 5 functions with different test cases during the run time of the program.

The code:

```
#include <iostream>
#include <cmath>
using namespace std;
int main(){
    int choice;

    cout << "Choose the function you would like to use: " << endl;
    cout << "1. Convert temperature from Celsius to Fahrenheit" << endl;
    cout << "2. Convert temperature from Fahrenheit to Celsius" << endl;
    cout << "3. Find the root of a quadratic equation ax² + bx + c = 0" << endl;
    cout << "4. Compute the BMI and determine whether the person's weight status" << endl;
    cout << "5. Calculate the sum from 1 to n except those that are divisible by 3" << endl;
    cout << "6. Exit" << endl;
    cout << "Enter your choice (1-6): ";
    cin >> choice;
    cout << endl;

    if (choice == 1) {
        float celTemp1;
        cout << "Enter the temperature in Celsius: ";
        cin >> celTemp1;
        cout << endl;
        float fahTemp1 = (celTemp1 * 9.0/5.0) + 32;
        cout << celTemp1 << " °C in Fahrenheit is: " << fahTemp1 << " °F" << endl;
    }
    else if (choice == 2 ){
        float fahTemp2;
        cout << "Enter the temperature in Fahrenheit: ";
        cin >> fahTemp2;
        cout << endl;
        float celTemp2 = (fahTemp2 - 32) * 5.0/9.0;
        cout << fahTemp2 << " °F in Celsius is: " << celTemp2 << " °C" << endl;
    }
```

```
    else if (choice == 3){
        int a, b, c;
        cout << "Enter the value of a, b and c into the quadratic equation: " << endl;
        cout << "Enter a: ";
        cin >> a;
        cout << "Enter b: ";
        cin >> b;
        cout << "Enter c: ";
        cin >> c;

        float discriminant = b*b - 4*a*c;

        if (discriminant > 0) {
            float root1 = (-b + sqrt(discriminant)) / (2 * a);
            float root2 = (-b - sqrt(discriminant)) / (2 * a);
            cout << "The roots are real and different: " << root1 << " and " << root2 << endl;
        }
        else if (discriminant == 0) {
            float root = -b / (2 * a);
            cout << "The roots are real and equal: " << root << endl;
        }
        else {
            cout << "The roots are complex and different" << endl;
        }
    }
    else if (choice == 4 ){
        float weight, height;
        cout << "Enter your weight (in kilograms): ";
        cin >> weight;
        cout << "Enter your height (in meters): ";
        cin >> height;

        float bmi = weight / (height * height);
        cout << "Your BMI is: " << bmi << endl;

        if (bmi < 18.5){
            cout << "You are underweight, you should eat more" << endl;
        }
        else if (bmi < 25){
            cout << "Your weight is the standard normal weight!" << endl;
        }
        else if (bmi < 30){
            cout << "You are overweight, you need to get some exercise in" << endl;
        }
        else {
            cout << "You are obese, that's concerning" << endl;
        }
    }
```

```
    else if (choice == 5 ){
        int n;
        cout << "Enter a positive integer n: ";
        cin >> n;

        int sum = 0;
        for (int i = 1; i <= n; i++) {
            if (i % 3 != 0) {
                sum += i;
            }
        }
        cout << "The sum from 1 to " << n << " except those that are divisible by 3 is: " << sum << endl;
    }
    else if (choice == 6 ){
        cout << "Exiting the program. Goodbye!" << endl;
    }
    else {
        cout << "Invalid choice. Please choose a number between 1 and 6." << endl;
    }


}
```

The output:

Choice 1:

```
Choose the function you would like to use:
1. Convert temperature from Celsius to Fahrenheit
2. Convert temperature from Fahrenheit to Celsius
3. Find the root of a quadratic equation ax² + bx + c = 0
4. Compute the BMI and determine whether the person's weight status
5. Calculate the sum from 1 to n except those that are divisible by 3
6. Exit
Enter your choice (1-6): 1

Enter the temperature in Celsius: 25

25 °C in Fahrenheit is: 77 °F
```

Choice 2:

```
Enter your choice (1-6): 2

Enter the temperature in Fahrenheit: 120

120 °F in Celsius is: 48.8889 °C
```

Choice 3:

```
Enter your choice (1-6): 3

Enter the value of a, b and c into the quadratic equation:
Enter a: 2
Enter b: 4
Enter c: 6
The roots are complex and different
```

Choice 4:

```
Enter your choice (1-6): 4

Enter your weight (in kilograms): 65
Enter your height (in meters): 1.73
Your BMI is: 21.7181
Your weight is the standard normal weight!
```

Choice 5:

```
Enter your choice (1-6): 5

Enter a positive integer n: 8
The sum from 1 to 8 except those that are divisible by 3 is: 27
```

Choice 6:

```
Enter your choice (1-6): 6

Exiting the program. Goodbye!
```